EEECE7352 Computer Architecture

Homework 4

Sakthivel P. Sivakumar

NUID: 002312294

Part E: (15 Points)

O0 Optimization Run Results: -

```
bash-4.4$ grep -i -e "L1 Instruction Cache" -e "L1 Data Cache" -e "L2 Unified Cache" -e "L3 Unified Cache" -e "Miss Rate" O0.out
    Load Miss Rate:      0.00%
   Store Miss Rate:       nan%
   Total Miss Rate:      0.00%
    Load Miss Rate:      4.44%
   Store Miss Rate:       nan%
   Total Miss Rate:      4.44%
 L1 Instruction Cache:
    Load Miss Rate:      0.00%
   Store Miss Rate:       nan%
   Total Miss Rate:      0.00%
 L1 Data Cache:
    Load Miss Rate:      6.26%
   Store Miss Rate:     16.67%
   Total Miss Rate:      8.86%
 L2 Unified Cache:
    Load Miss Rate:     94.20%
   Store Miss Rate:      6.26%
   Total Miss Rate:     52.87%
 L3 Unified Cache:
    Load Miss Rate:      0.08%
   Store Miss Rate:    100.00%
   Total Miss Rate:      5.64%
```

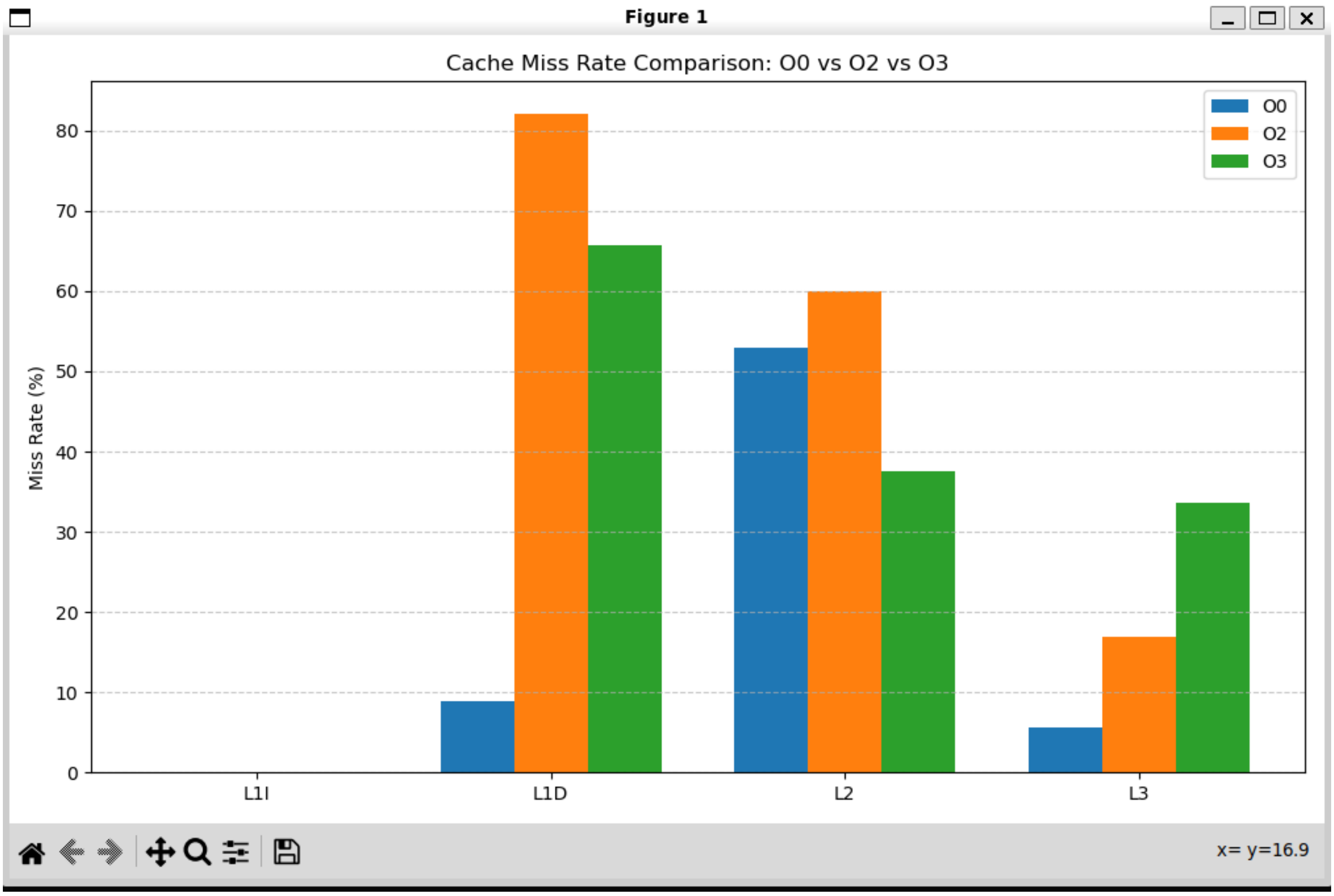
O2 Optimization Run Results: -

```
bash-4.4$ grep -i -e "L1 Instruction Cache" -e "L1 Data Cache" -e "L2 Unified Cache" -e "L3 Unified Cache" -e "Miss Rate" O2.out
    Load Miss Rate:      0.00%
   Store Miss Rate:       nan%
   Total Miss Rate:      0.00%
    Load Miss Rate:     32.97%
   Store Miss Rate:       nan%
   Total Miss Rate:     32.97%
 L1 Instruction Cache:
    Load Miss Rate:      0.01%
   Store Miss Rate:       nan%
   Total Miss Rate:      0.01%
 L1 Data Cache:
    Load Miss Rate:     73.57%
   Store Miss Rate:     99.11%
   Total Miss Rate:     82.03%
 L2 Unified Cache:
    Load Miss Rate:     83.20%
   Store Miss Rate:     24.97%
   Total Miss Rate:     59.91%
 L3 Unified Cache:
    Load Miss Rate:      0.27%
   Store Miss Rate:    100.00%
   Total Miss Rate:     16.89%
```

O3 Optimization Run Results: -

```
bash-4.4$ grep -i -e "L1 Instruction Cache" -e "L1 Data Cache" -e "L2 Unified Cache" -e "L3 Unified Cache" -e "Miss Rate" O3.out
    Load Miss Rate:       0.00%
   Store Miss Rate:        nan%
   Total Miss Rate:       0.00%
    Load Miss Rate:       0.39%
   Store Miss Rate:        nan%
   Total Miss Rate:       0.39%
 L1 Instruction Cache:
    Load Miss Rate:       0.01%
   Store Miss Rate:        nan%
   Total Miss Rate:       0.01%
 L1 Data Cache:
    Load Miss Rate:      49.12%
   Store Miss Rate:      99.10%
   Total Miss Rate:      65.67%
 L2 Unified Cache:
    Load Miss Rate:      50.01%
   Store Miss Rate:      24.97%
   Total Miss Rate:      37.51%
 L3 Unified Cache:
    Load Miss Rate:       0.68%
   Store Miss Rate:     100.00%
   Total Miss Rate:      33.68%
```

Cache Miss rate Comparison Plot



In this experiment, I used the Pin tool and the allcache.so instrumentation to measure the cache behaviour of a simple array-traversal program compiled with different optimization levels. I collected miss-rate data for multiple cache levels (L1 Instruction, L1 Data, L2 Unified, and L3 Unified) and plotted the results for comparison.

- **L1 Instruction Cache (L1I)**

  L1I miss rates remain extremely low across all optimization levels (0.00 - 0.01%). Although -O2 and -O3 perform loop transformations (unrolling, strength reduction, dead-code elimination), these optimizations reduce the number of dynamic instructions and do not significantly enlarge the binary enough to cause I-cache pressure. As a result, L1I behaviour is stable and nearly identical across all optimization levels.

- **L1 Data Cache (L1D)**

  L1D miss rates are very high for all three versions, with large jumps for the -O2 binary (82%). This change is because the column-major traversal has extremely poor spatial locality. Compiler optimizations cannot fix the stride access pattern, so L1D misses remain dominated by this loop. The differences between O0, O2, and O3 occur due to loop restructuring and unrolling, but the underlying memory pattern is still the main cause of L1D misses.

- **L2 Unified Cache**

  L2 also shows high miss rates because most L1D misses propagate to L2, and the large array easily exceeds L2 capacity. O3 tends to reduce L2 misses slightly due to better loop unrolling and scheduling, while O2 increases pressure by issuing memory accesses more tightly. Overall, L2 behaviour mainly reflects the poor locality of the column-major phase.

- **L3 Unified Cache (LLC)**

  LLC miss rates vary with optimization level. O0 issues memory requests more slowly and spreads out accesses, causing fewer conflicts in L3. O2 and O3 generate tighter loops and more back-to-back loads, increasing pressure on the LLC. Thus, L3 misses are influ enced more by compiler scheduling than by locality.